

TRAFFIC LIGHT

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I. Problem

The traffic light system contains the following items:

- The START, STOP buttons; The DAY-NIGHT switch.
- 03 pairs of W-E lights: GR1, YL1, RD1; 03 pairs of N-S lights: GR2, YL2, RD2.

The process:

- Select the DAY-NIGHT mode using the switch.
- Press the START button to start the system.
- The DAY mode - The system works automatically according to the following cycle length: (*) GR1/RD2: 12s → YL1/RD2: 3s → RD1/GR2: 12s → RD1/YL2: 3s → (*).
- The NIGHT mode - The system works after 10:00 p.m. according to the following cycle length: (**) YL1/YL2: 3s → NO_SIGNAL → (**).
- Press the STOP button to stop the system immediately.

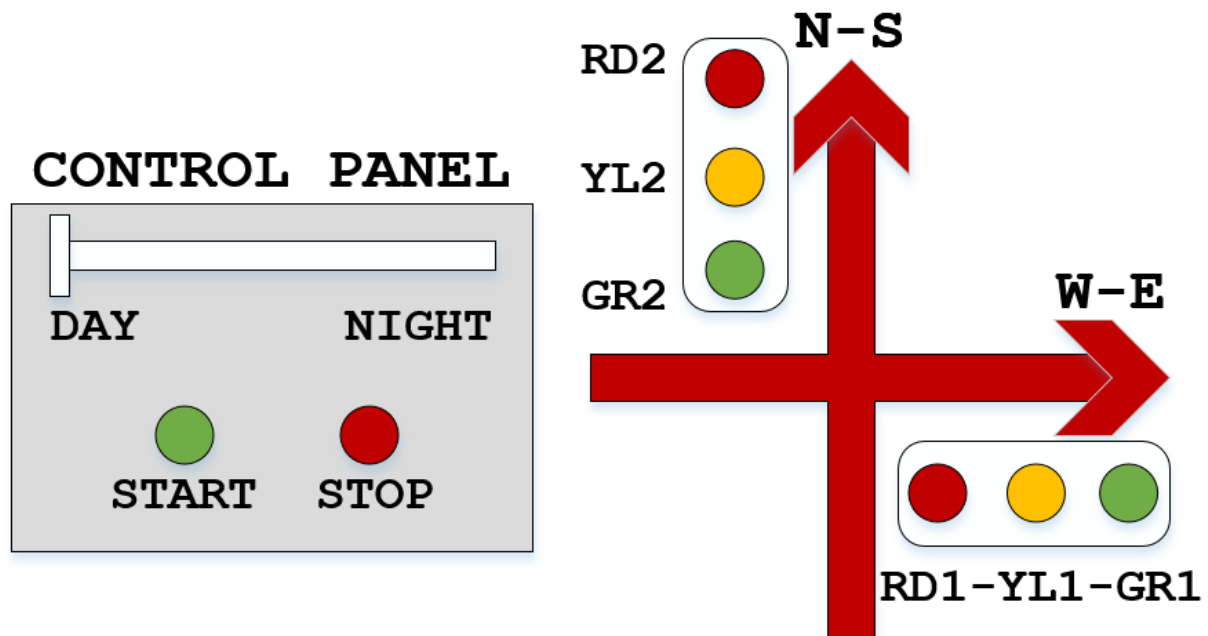


Figure 1.1. The traffic light system

II. PLC tags

INPUT		OUTPUT	
PIN	MEANING	PIN	MEANING
I0.0	START	Q0.0	RD1
I0.1	STOP	Q0.1	YL1
I0.2	DAY	Q0.2	GR1
I0.3	NIGHT	Q0.3	RD2
		Q0.4	YL2
		Q0.5	GR2

III. Grafcet

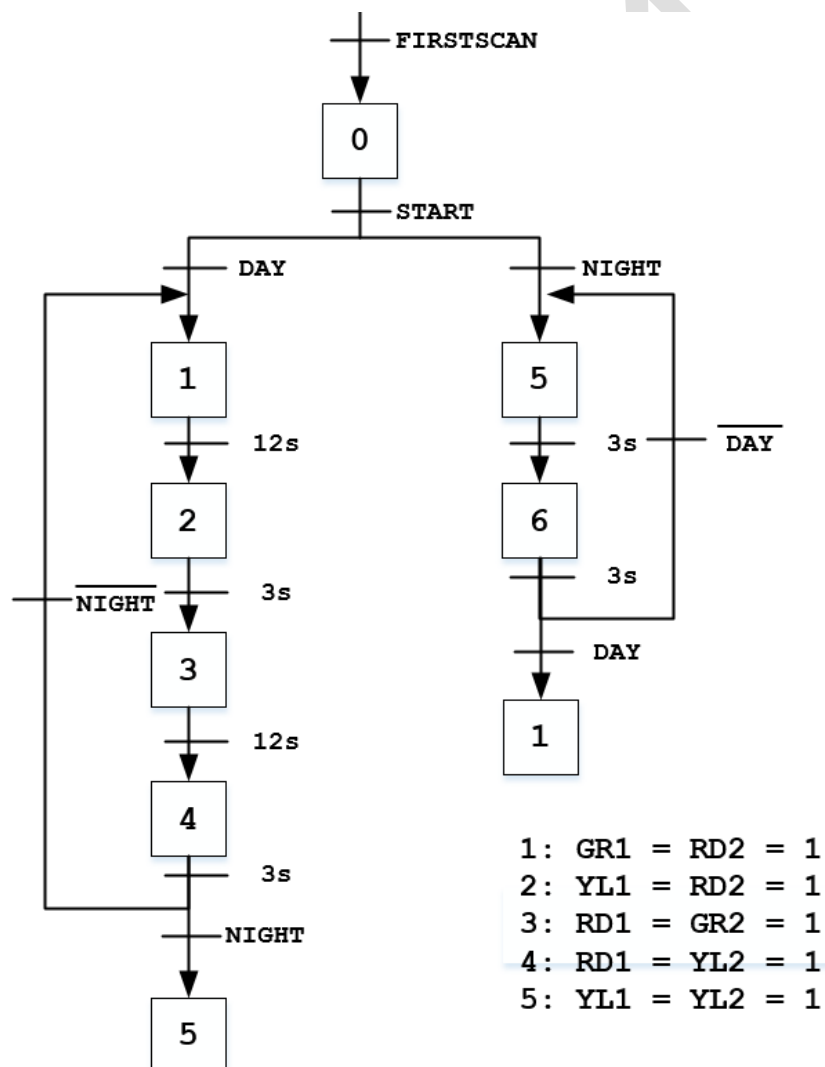


Figure 2.1. The Grafcet

IV. Program

See the attached file (TIA Portal v14 SP1).